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Assignment & Course Details:

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Course *(e.g. Bachelor in Computing)* : **Bachelor of Computer Science (Hons)**	
Lecturer Name: Mr Julian Wong Choon Yee	

Assessment Due Date: *(dd/mm/yy)*	17/7/2025	**Assessment Title:**	Internet and Web Development Group Assignment Proposal

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Project Proposal: Student Performance Monitoring System (SPMS) for Programme Leaders

1.0 Introduction

In higher education institutions, Programme Leaders (PLs) responsible for monitoring student performance and academic progression. However, manual tracking systems are inefficient, error-prone, and do not support real-time information. The lack of automated monitoring technologies limits the PL's ability to detect at-risk students and make data-driven academic planning decisions.

1.1 Purpose and Goals of the Website

The proposed web-based Student Performance Monitoring System (SPMS) aims to provide university program leaders with a complete, real-time platform for monitoring, analyzing and managing student academic performance. The primary purpose is to provide data-driven insights to support early identification of students at risk, optimize academic planning, and improve student retention rates. This system will facilitate tracking indicators like Cumulative Grade Point Average (CGPA), credit accumulation, progression status, academic workload, and module completion. It will also include visualization features to help clarify complex academic data and improve decision-making.

1.2 Identification of the Target Audience and Their Needs

The primary target audience for this system includes:

1. Programme Leaders
    - Require clear academic data dashboards to track student performance.
    - Need alerts or insights to detect underperforming students early.
    - Prefer simple data filtering tools.
    - Must manage and update student-module records easily.
2. Academic Advisors / Lecturers

- Wants to track student progress in modules they oversee.
- Need performance summaries by cohort, module, or semester.

3. Administrative Staff

- Will input and manage data in bulk.
- Need access to audit logs or change histories for accountability.

1.3 Identification of the Programme Leader and Their Needs

Programme Leader Persona for Student Performance Monitoring System

Name: Miss Siti

Age: 42

Position: SCCM Programme Leader

Background: With over 10 years of academic and administrative experience, Miss Siti oversees 200+ students across multiple intakes and academic years. She coordinates with lecturers and the dean's office on student progression matters.

Goals & Responsibilities:

- Monitor student performance trends and identify those at risk of failing or delaying graduation.
- Ensure students fulfill graduation requirements (e.g., minimum CGPA, credits).
- Plan interventions or academic advice sessions based on performance data.
- Report performance summaries to the faculty board and QA committees.

Dissatisfaction of Existing System:

- Lack of visual indicators or automated alerts for students falling below thresholds.
- Time-consuming to retrieve and compare records for individual students across semesters.

Needs from the System:

- A centralized dashboard with filtering options (e.g., by semester, risk level).
- Early warning indicators (e.g., CGPA < 2.00, incomplete modules).
- Exportable reports for meetings.
- Secure access with minimal administrative overhead.

1.4 Problem Statement and Justification

1.4.1 Problem Statement

In many higher education institutions, Programme Leaders rely heavily on fragmented or outdated student records that are often scattered across spreadsheets, student information system (SIS) platforms, and offline documentation. Despite ongoing digital transformation efforts, existing systems fail to offer actionable insights into student progression or academic risk. As a result, Programme Leaders struggle with identifying students who are off-track, planning interventions, and ensuring timely graduation.

1.4.2 Justification for Why this Website Is Needed

A dedicated web-based Student Performance Monitoring System (SPMS) is essential to bridge the gap between raw academic data and strategic academic decision-making. Unlike traditional SIS platforms, a custom-built web system tailored to the needs of Programme Leaders can:

- Centralize and standardize student performance data.
- Offer interactive dashboards and reporting tools.
- Support predictive analytics for student retention and success.
- Improve responsiveness to academic challenges.

The integration of data analytics in education has been proven to significantly improve academic outcomes (Siemens & Long, 2011). Moreover, web-based platforms enhance accessibility and ease of use, enabling Programme Leaders to act quickly on emerging academic issues (Chatti et al., 2012).

2.0 Functionalities and Specifications

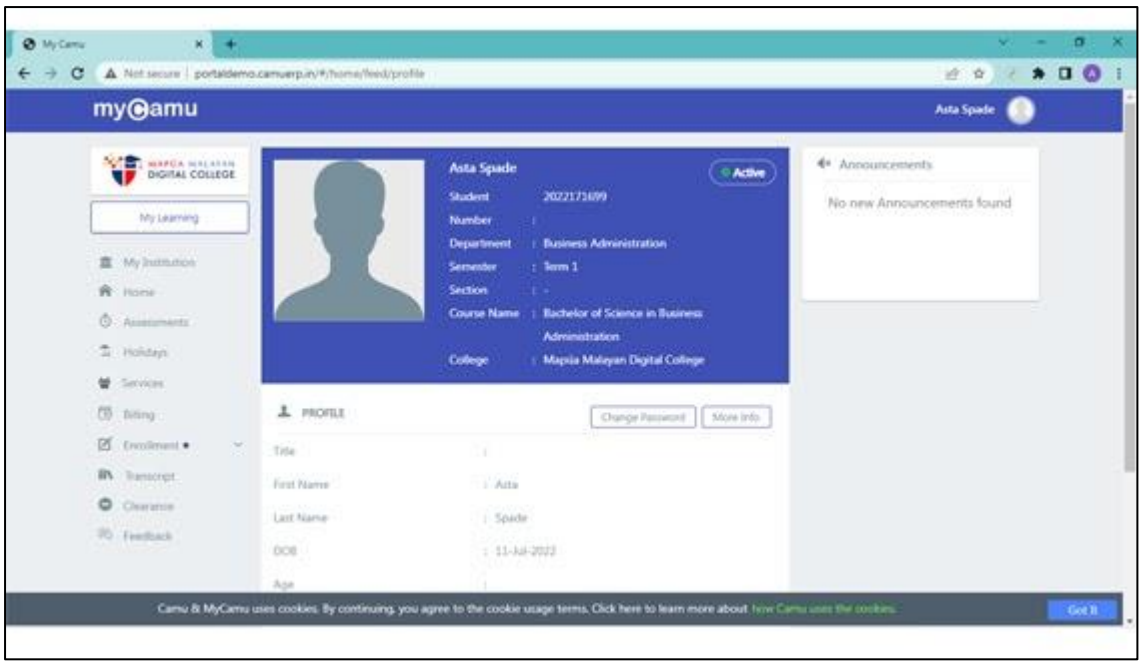
2.1 Similar Apps

As student performance monitoring system was widely used in most of universities, it was designed and developed differently according to the needs of each university. For example, MyCamu, Teach n' go, and Education Horizon: Zunia system were served with same generic function of student performance monitoring system for university program leader to monitor student performance. This section compares the three platforms mentioned above in detail, along with a set of comparison criteria listed below to get a better view of functionalities a great student performance monitoring system.

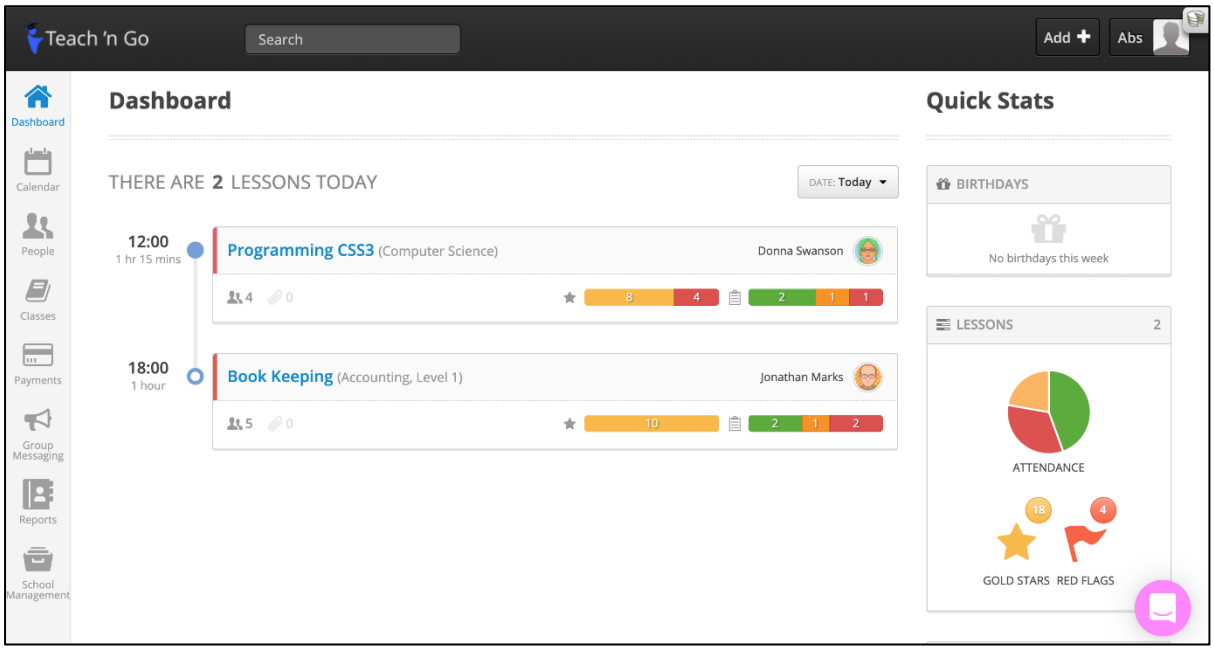
The comparison criteria of the selected websites include:

1. User Experience – Ease of use for Program Leader
2. Functionality – Functions that Program Leader that able to use and access
3. Pricing – Accessible for free or require paid subscriptions

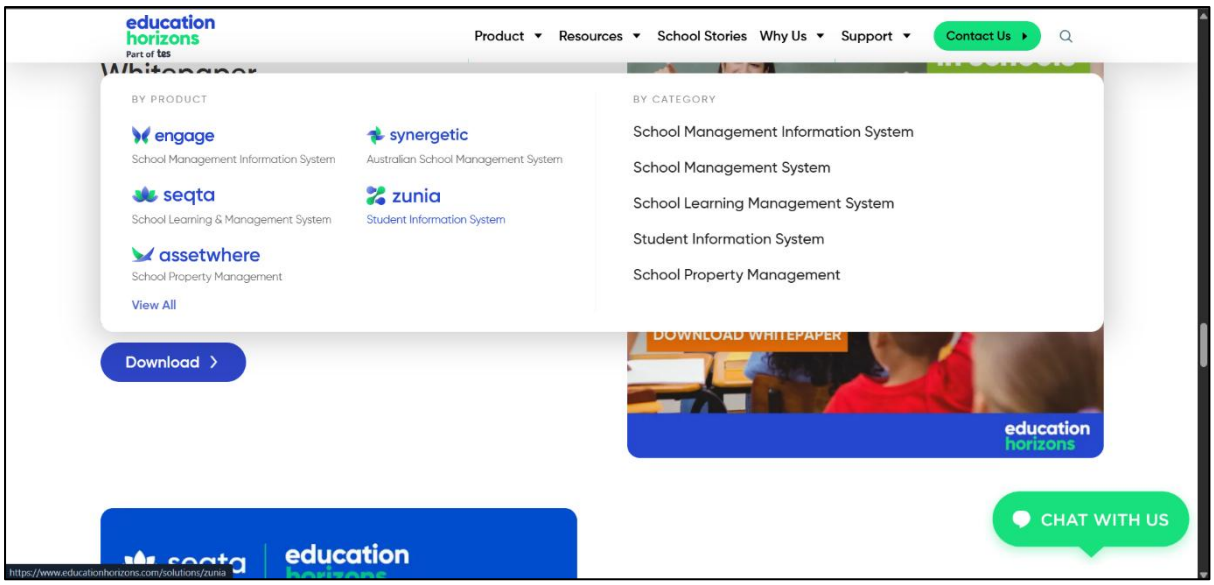
Feature	MyCamu	Teach n' go	Education Horizon: Zunia
UI/UX	Functional & smooth performance, with simple design	Functional & smooth performance with modern UI design	Functional & smooth performance with modern UI design
Functionality	Basic functions for PL to use and access	Modern functions with simple design for PL to use and understand	Advanced functions and understandable design for PL
Pricing Subscription	Accessible for free but require logging in with specific email	Require paid subscriptions with provide 14-day free trial to use	Require paid subscriptions after contact to the official.



This picture shows the UI of MyCamu from one of the schools, the design includes basic functions of the system such as view student list and profile, course details, transcript and billing information and more. The design was simple to understand and to use and access as the categories of each function is clearly stated and PL able to navigate into the category they want.



Teach 'n go, with almost the same functionality as MyCamu, but with more detailed category included such as student's other personal information, direct messages to students are available, academic calendar and more. The design is simple and understandable to use and access, which almost the same as MyCamu.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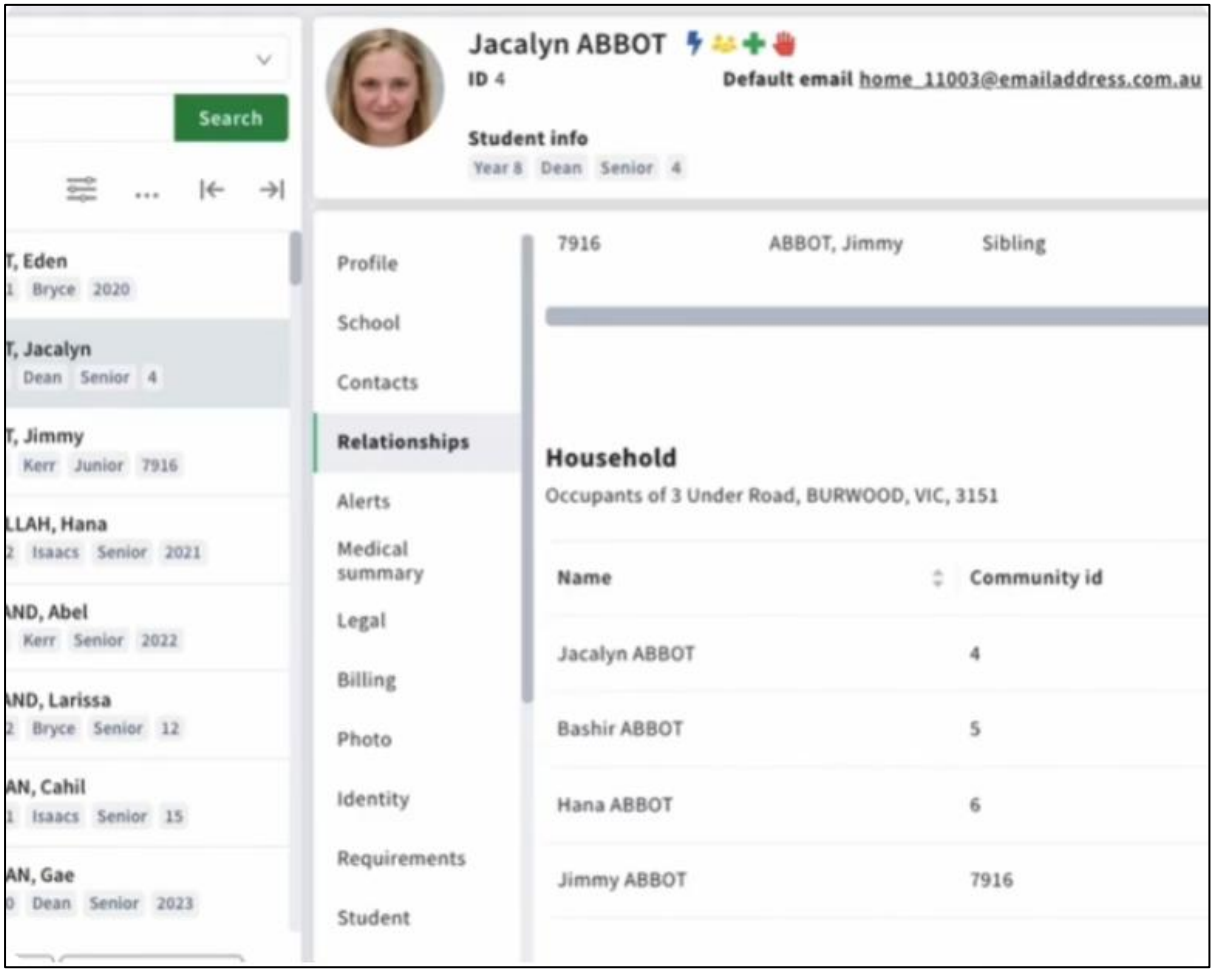
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0 Dean Senior 2023
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ID 4
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Bashir ABBOT
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Hana ABBOT
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Jimmy ABBOT
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Education Horizon is a special company that provides education system to school according to the products they provide in the 1st picture with different category. Zuria, one of the products they provided focused on managing students' information system. Same as MyCamu and Teach ‘n go, the design is smoother and with more detailed include direct parent communication function, detailed information about student, general ledger integrations and more. This student information system suitable for school nowadays as the function is professional enough that it have most of the functionality to access for PL.

2.2 Functional Specifications

The table below shows the main functional specifications of Student Performance Monitoring System (SPMS).

ID	Function	Description
F001	User Authentication	The system shall allow users to login securely using their credentials.
F002	Student Management	The system shall allow users to add, edit, delete, and bulk import student records via CSV files.
F003	Course & Module Management	The system shall allow users to create, edit, and delete courses and modules.
F004	Assign Students	The system shall allow users to assign students to specific courses or programme tracks.
F005	Manage Module Status	The system shall allow users to mark modules as completed or exempted.
F006	Database Integrity	The system shall enforce foreign key constraints and maintain referential integrity.
F007	Data Backup & Recovery	The system shall support data backup and recovery features.
F008	Duplicate Detection	The system shall detect and handle duplicate student records during data entry or import.
F009	Student Profile Dashboard	The system shall display student personal and academic information, including name, ID, programme details.
F010	Academic Record Display	The system shall show completed subjects with subject code, name, grade, credits, and CGPA, organized by semester.
F011	Exempted & Transferred Modules	The system shall list modules that are exempted or credit transferred.
F012	Upcoming Subjects	The system shall highlight subjects planned for the upcoming semester.
F013	Data Visualization (Student)	The system shall provide visual charts for CGPA trends, pass/fail ratios, and subject averages on the student dashboard.

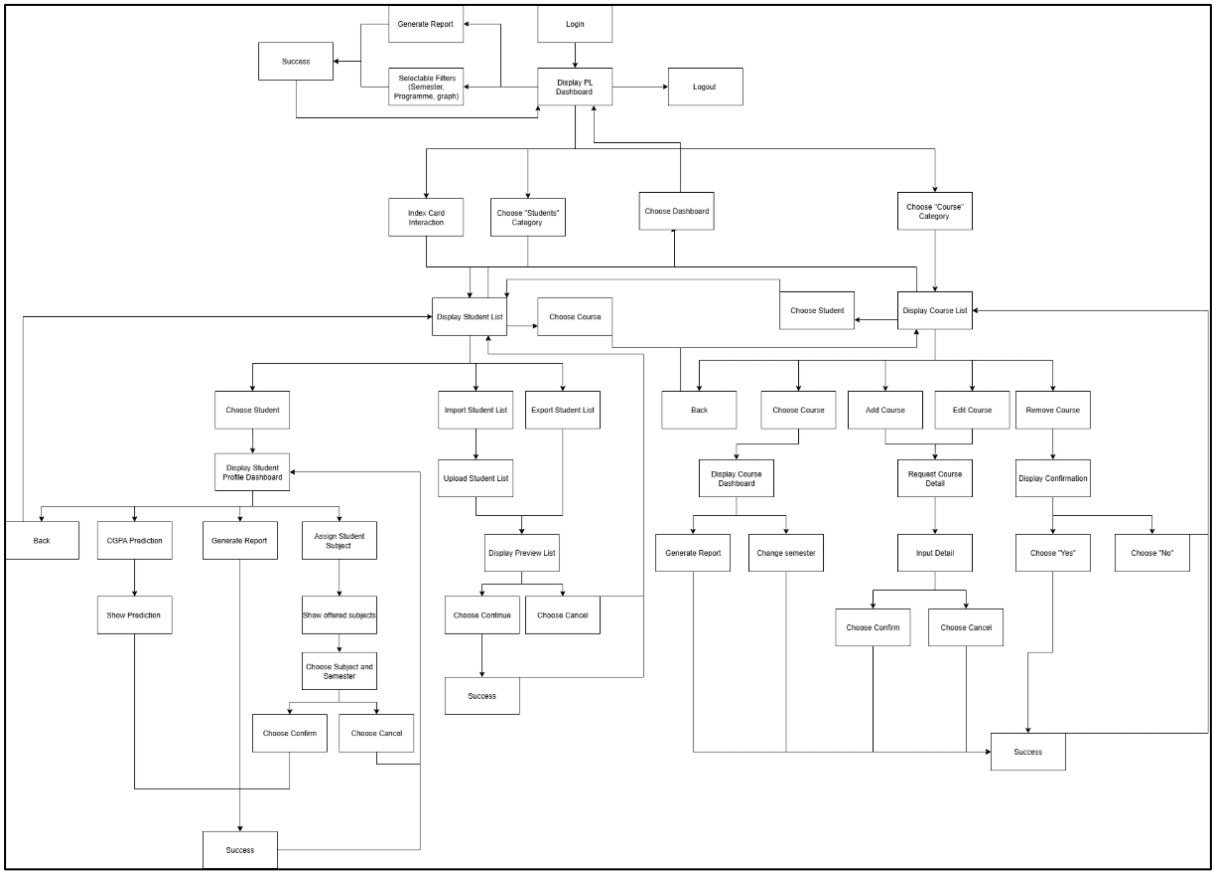
F014	PL Dashboard	The system shall provide program leaders with dashboards showing overall student performance indicators such as Dean's List and Probation List.
F015	Module Breakdown	The system shall display breakdowns of modules for the current semester, next semester, and remaining subjects for program leaders.
F016	Summary Reports	The system shall generate summary reports such as unregistered subjects and at-risk students for program leaders.
F017	Filtering & Sorting	The system shall allow filtering and sorting of data by student, semester, course, grade, etc. to improve decision-making.
F018	Semester-wise Overview	The system shall allow users to view semester-wise progression overviews.
F019	Interactive Dashboards	The system shall allow users to drill down into individual student or subject performance through interactive dashboards.
F020	Export Function	The system shall support exporting dashboard views and reports to CSV or PDF formats.
F021	Data Storytelling	The system shall automatically generate narrative summaries highlighting trends, warnings, and insights in plain language.
F022	Intuitive UI	The system shall provide an intuitive and user-friendly interface for all users.

2.3 Innovative Features

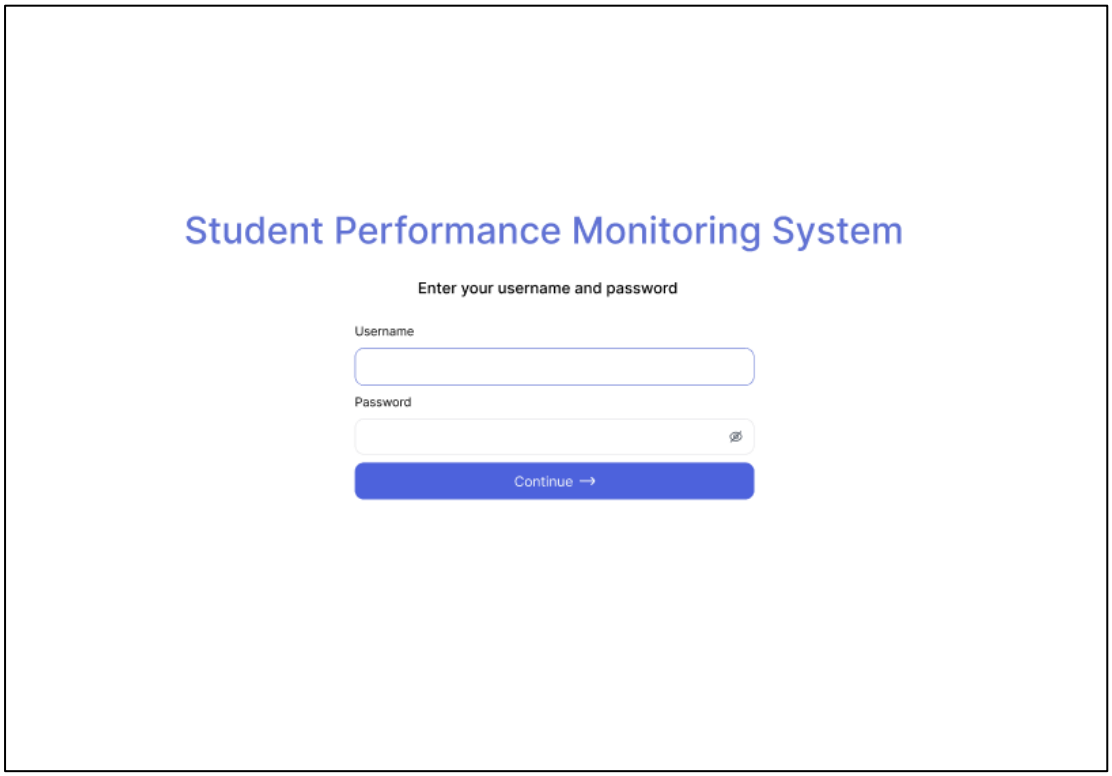
F023	Code Scanning	The system shall user to scan student ID barcode to retrieve the corresponding student's profile.
F024	CGPA Prediction	The system shall predict the students' performance (CGPA) based on their results.
F025	AI Report Generator	The system shall automatically summarize and generate report for pl dashboard, students' performance and course dashboard.

3.0 Wireframe

3.1 User Flow

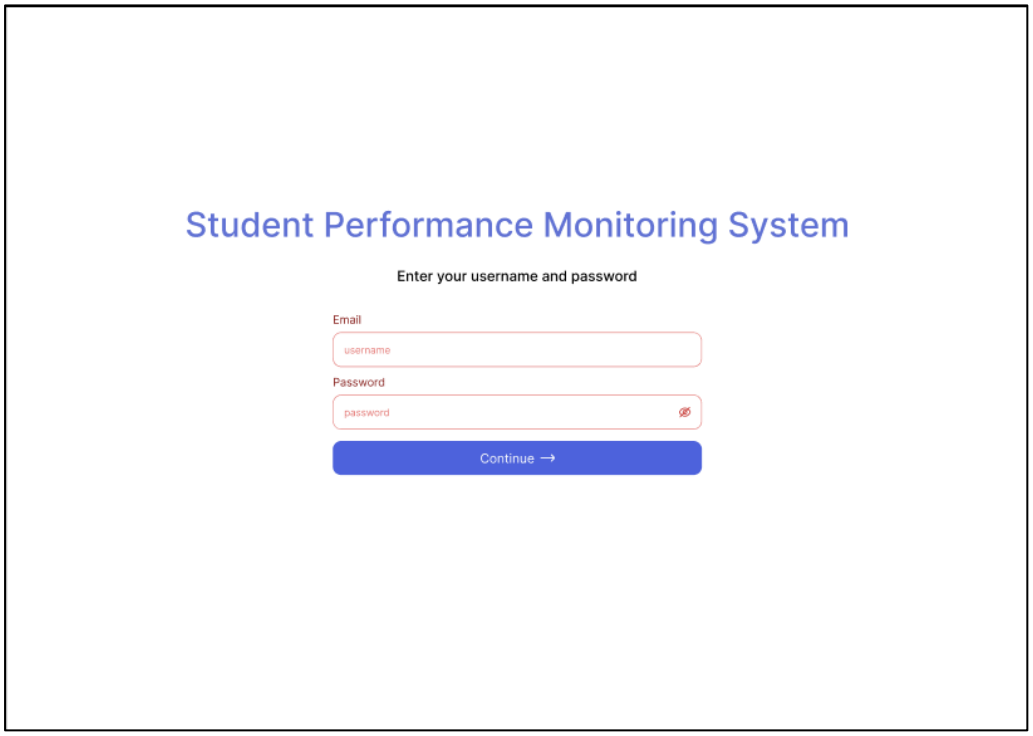


3.2 Login Page

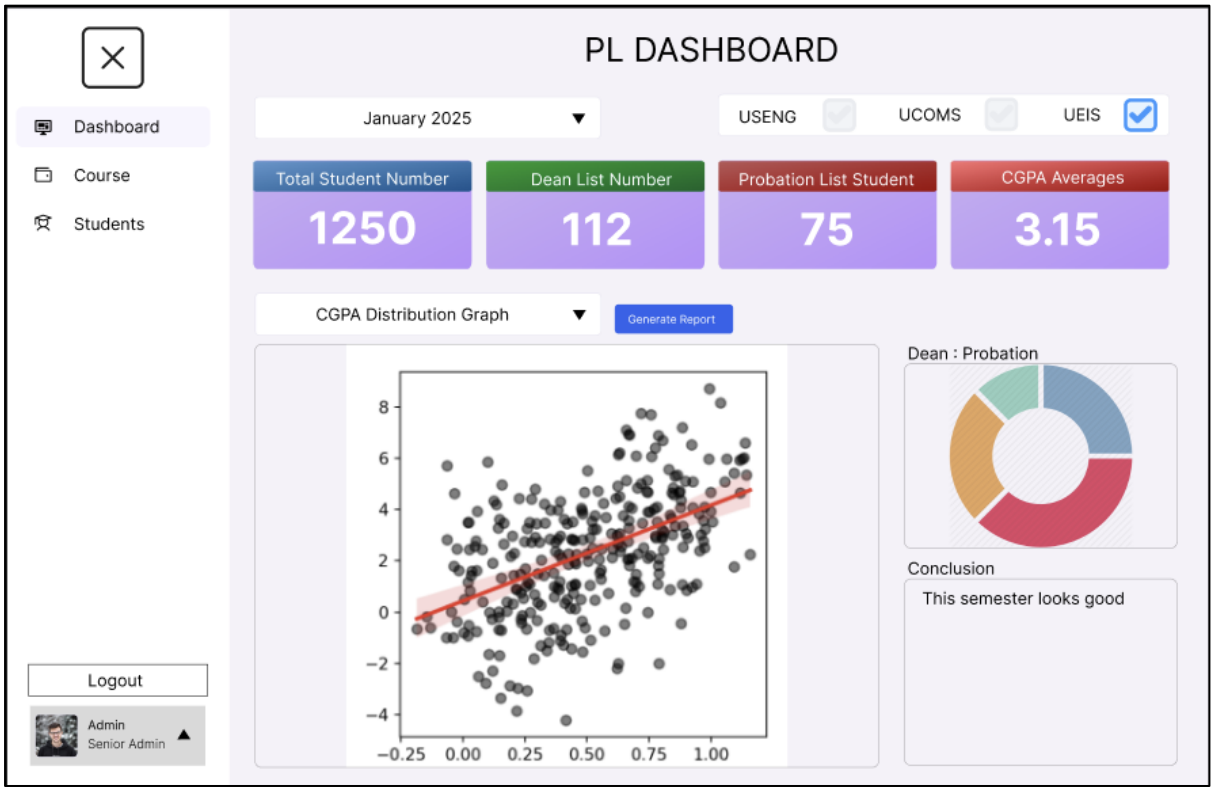


The figure above shows login page of the student performance monitoring system. User need input the correct username and password to access the system. If the username or password incorrect, system will show the error response and prompt user to input again.

The figure below shows the login page after input wrong username or password.

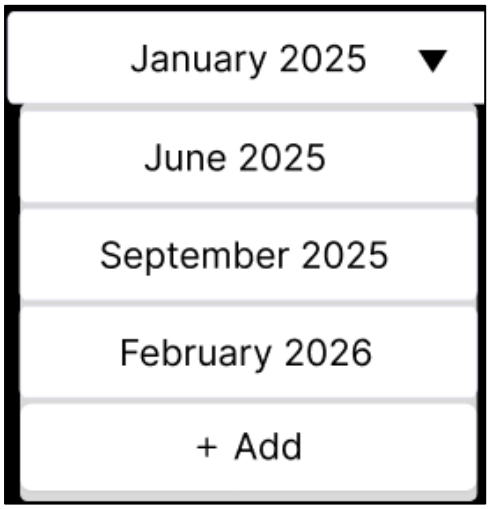


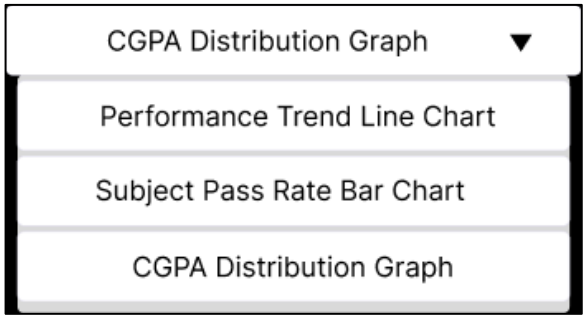
3.3 Main Page (Program Leader Dashboard)



Based on the figure above, the system will redirect the user to the main page once the user login is successful. The system shows the general overview of the students' performance such as the total number of students, the number of dean list student and probation student, the ratio of dean and probation students, student CGPA distribution graph, subject pass rate bar chart, performance trend line chart and the overall conclusion of the dashboard. In this page, the user can select a graph for data visualizations of student performance by choosing a semester with a dropdown selection and the programme with checkboxes for an interactive dashboard.

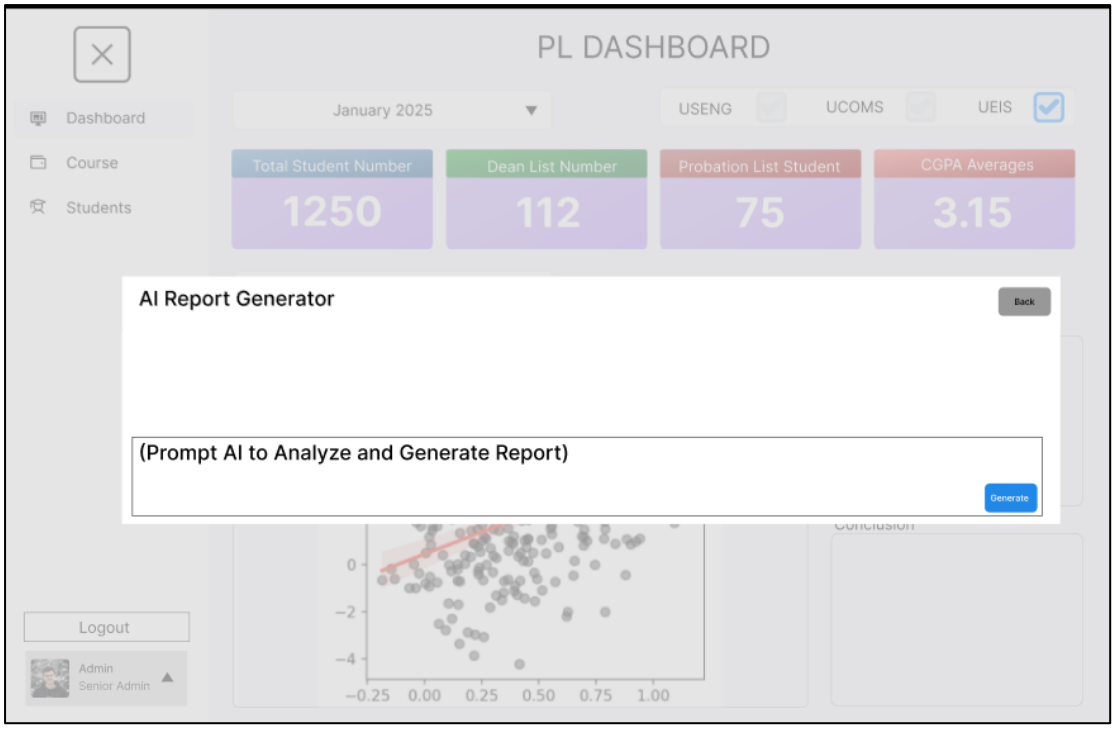
In Dropbox selection, system will provide the few options for user to choose. In addition, system also have provided the “+ Add” button to allow user to add the new semester manually. The figures below show the option included in the Dropbox.



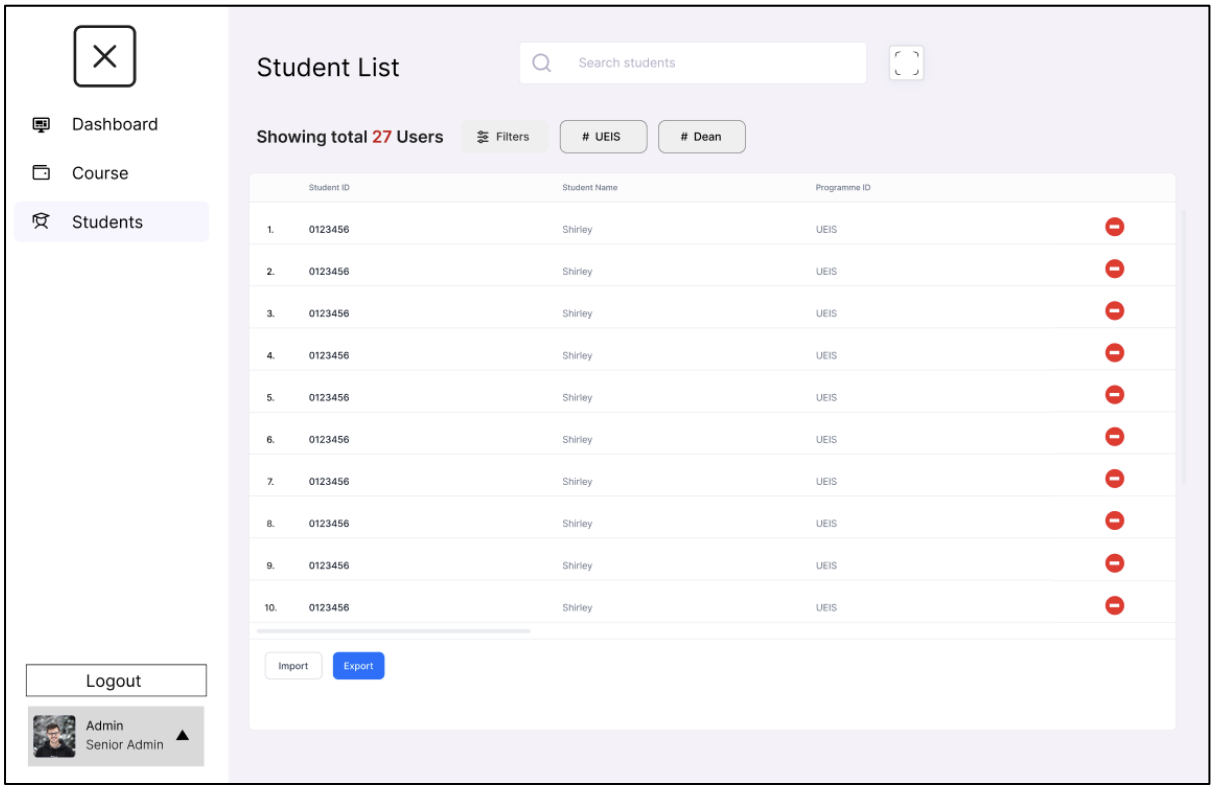


The index card can be interactive with the user as the system will redirect user to student list page with the specific filtering tag once the user clicked the index card needed. For example, selecting the probation list student index card will direct the user to the student list page with the probation filter tag.

Furthermore, system also provides an ai report generation function that allow users to prompt and generate report for the dashboard.

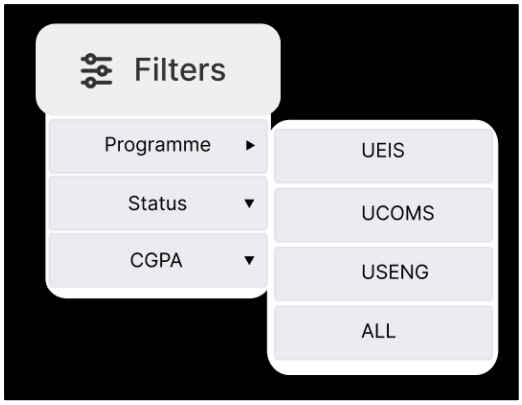


3.4 Student List Page



The figure above shows the student list page. After clicking a specific student, the system will direct the user to the student profile page.

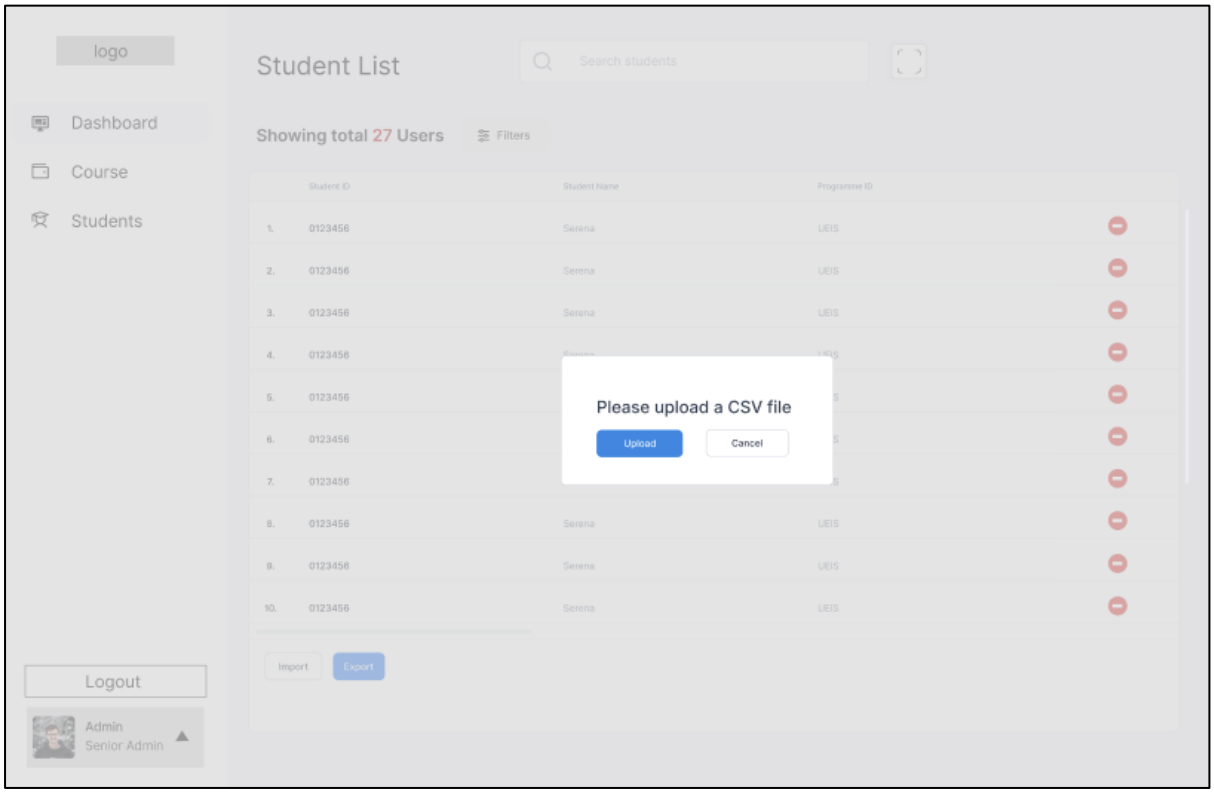
For the PL to filter the student list, the button “Filters” enables user to choose the filter condition based on programme, CGPA range and status of the student. The figure below shown the example of the button filtering function.

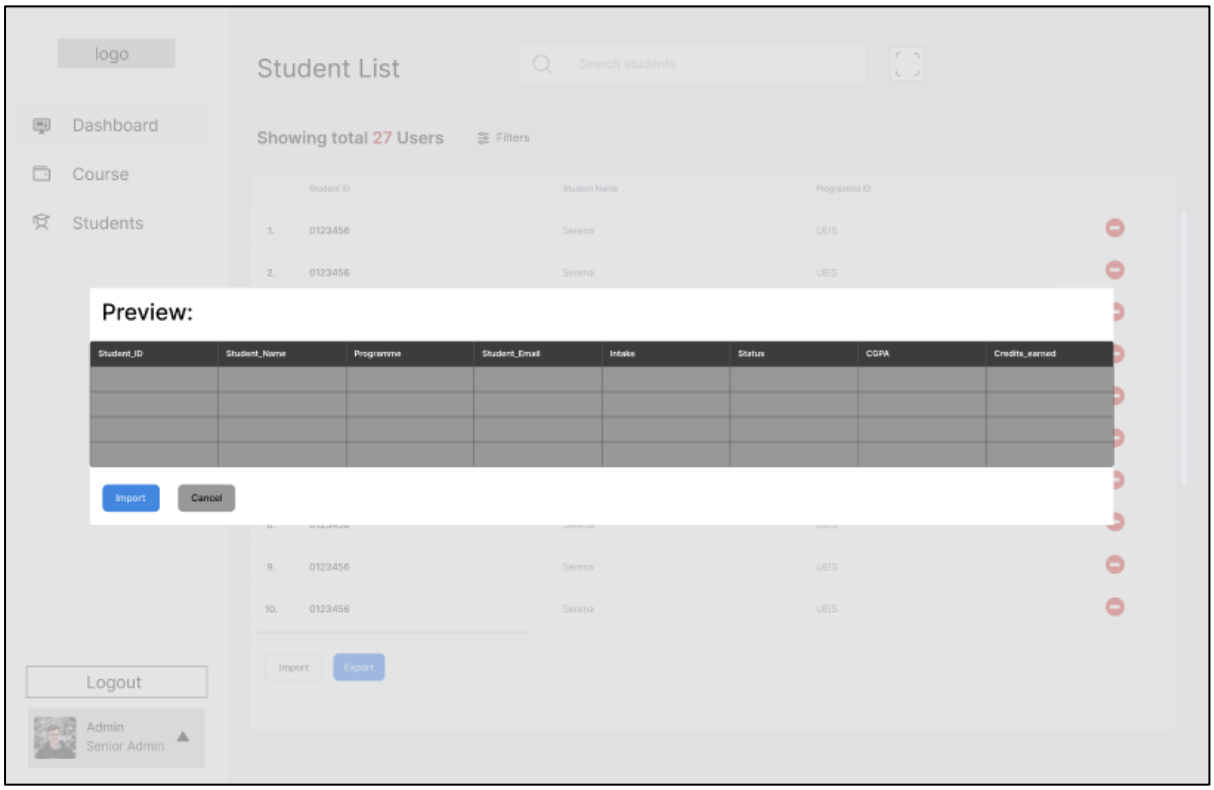


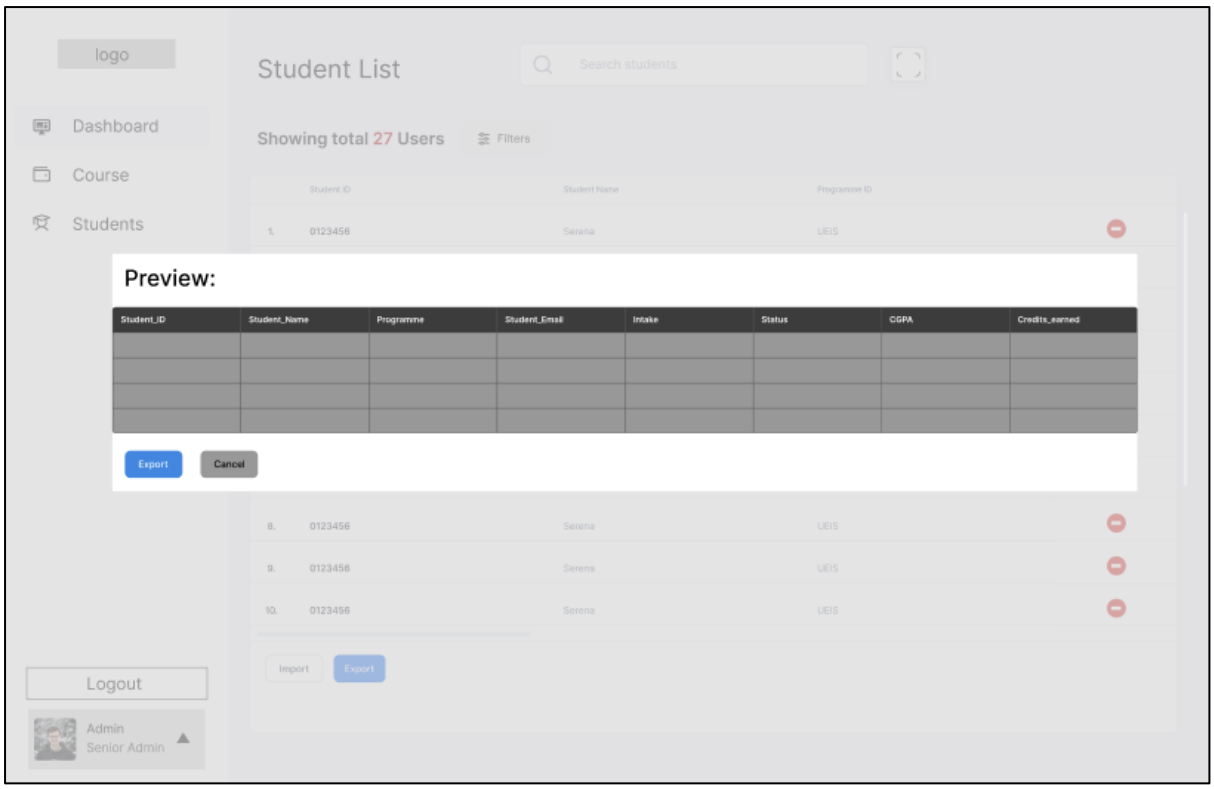
After the user chosen the filtering condition, this page will show the filtered data table and left the filter condition tag beside the filter button.

Besides that, the system also has a search bar for the user to input the text to find the required data such as student name and student id. Furthermore, the scanning button that located besides the blank can used to directly find the student through scanning the student barcode.

This page also allows the user to import the data in CSV format through clicking “import” button and export the CSV file from the student list table with the “export” button. Before the system imports or exports the data, the system will display a preview of the data before the user to confirms the import/export action.



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Student List
Search students
Dashboard
Course
Students
Showing total 27 Users
Filters
Preview:
Student_ID
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Credits_earned
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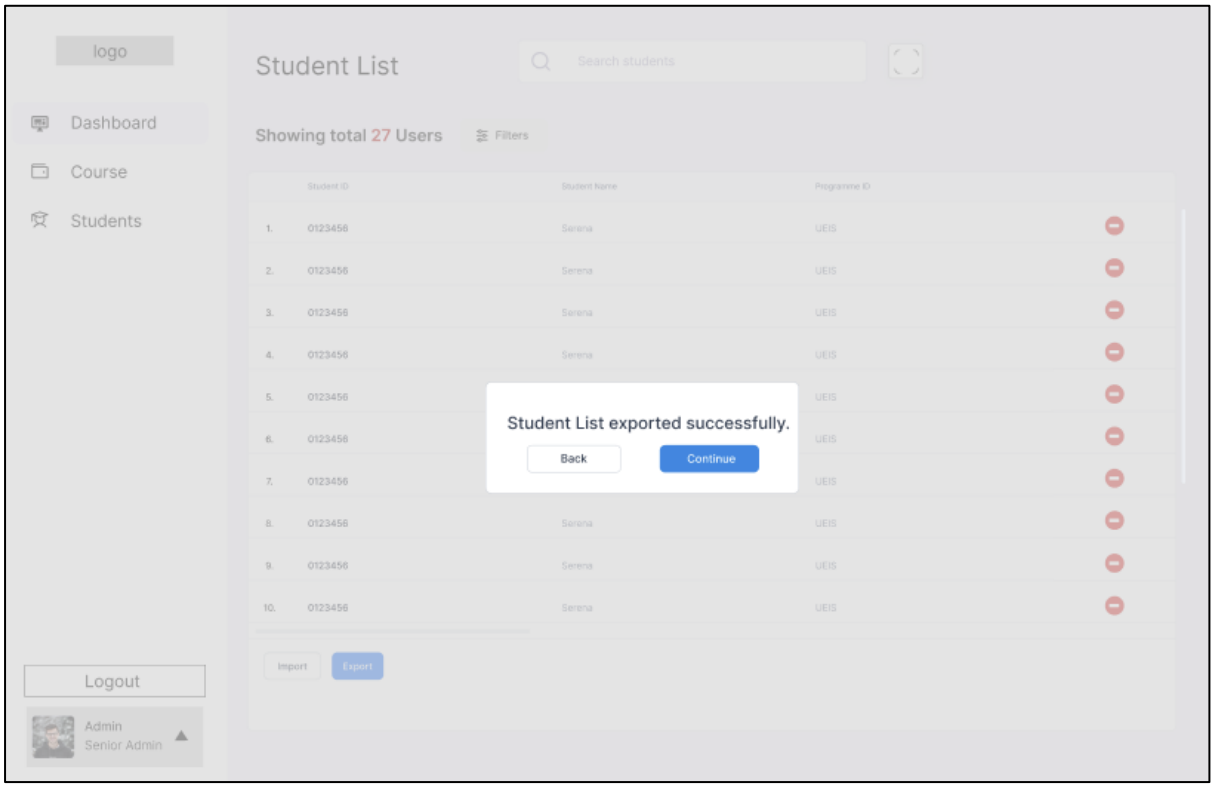
Logout
Admin
Senior Admin


logo
Student List
Search students
Dashboard
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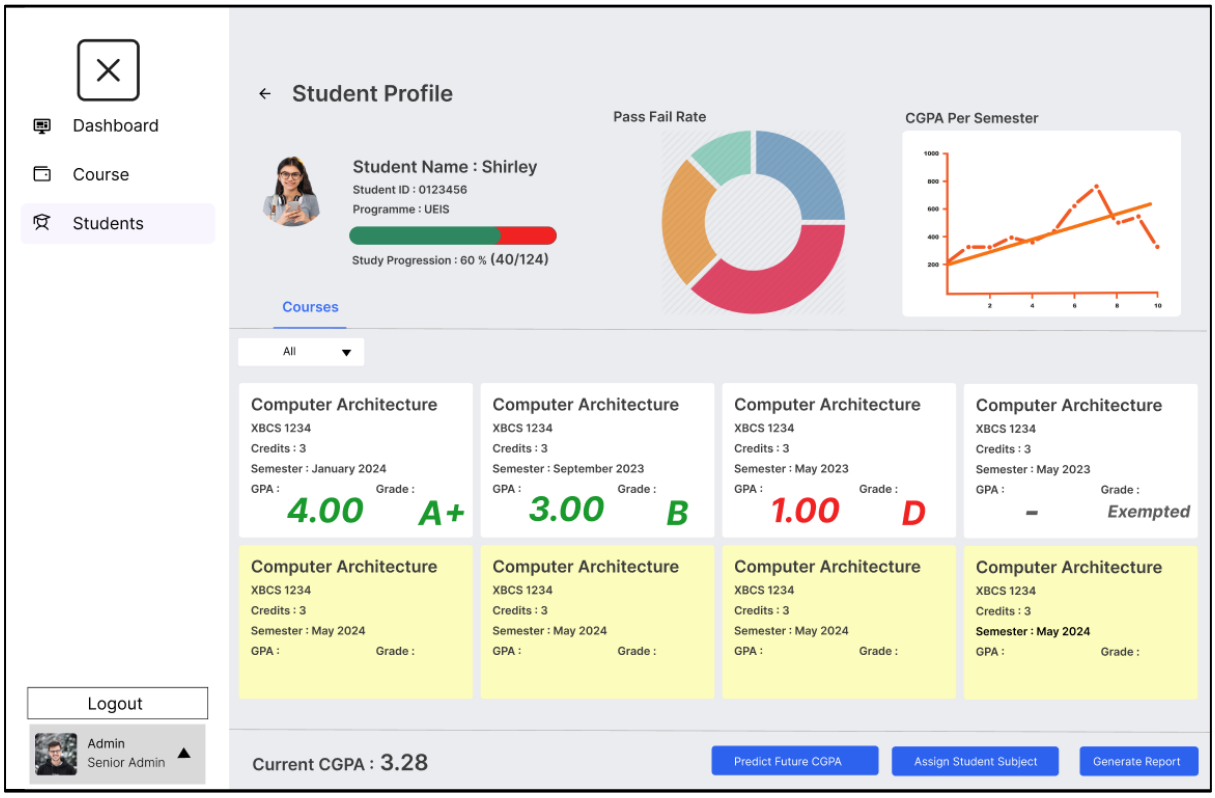
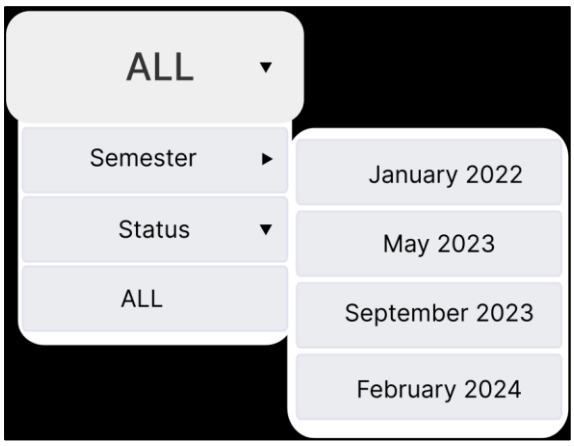
Students
Showing total 27 Users
Filters
Student List exported successfully.
Back
Continue
Logout
Admin
Senior Admin

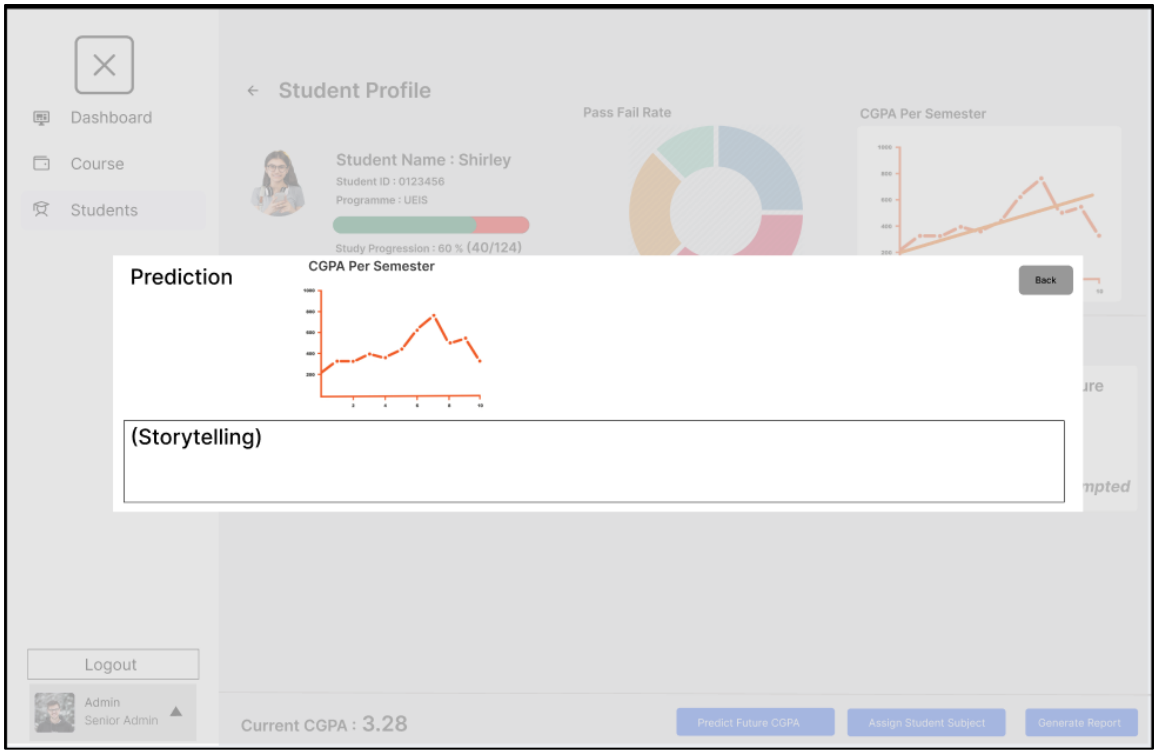
3.5 Student Profile Page



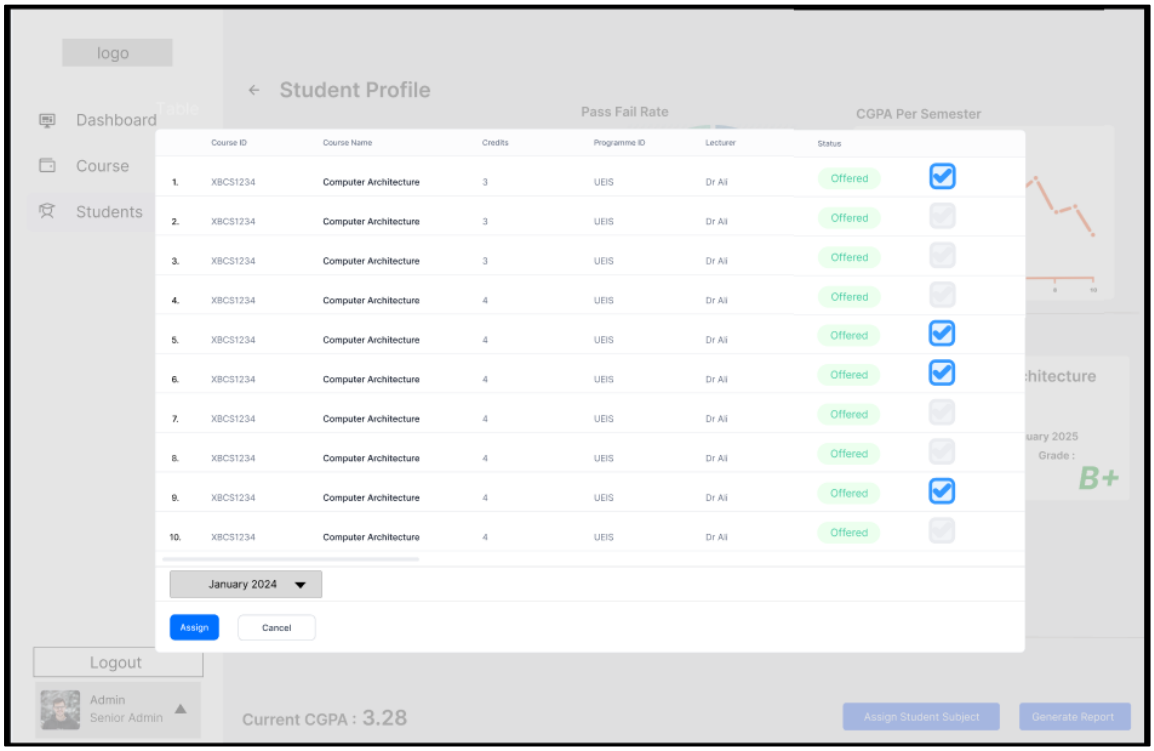
The figure above shows the student profile page where it displays the student profile dashboard. This page displays a student's general information such as student name, student id and programme. This page also displays data visualizations such as Student Pass-Fail Rate donut chart, CGPA Per Semester line chart, study progression bar. It also displays all completed and exempted subjects by the student including subject code, subject name, grade, credit, and CGPA organized by semester. The yellow highlighted subjects are the future subjects planned semesters. In addition, the user can filter the courses information based on requirements, the figure below shows the details of the dropbox in this page.



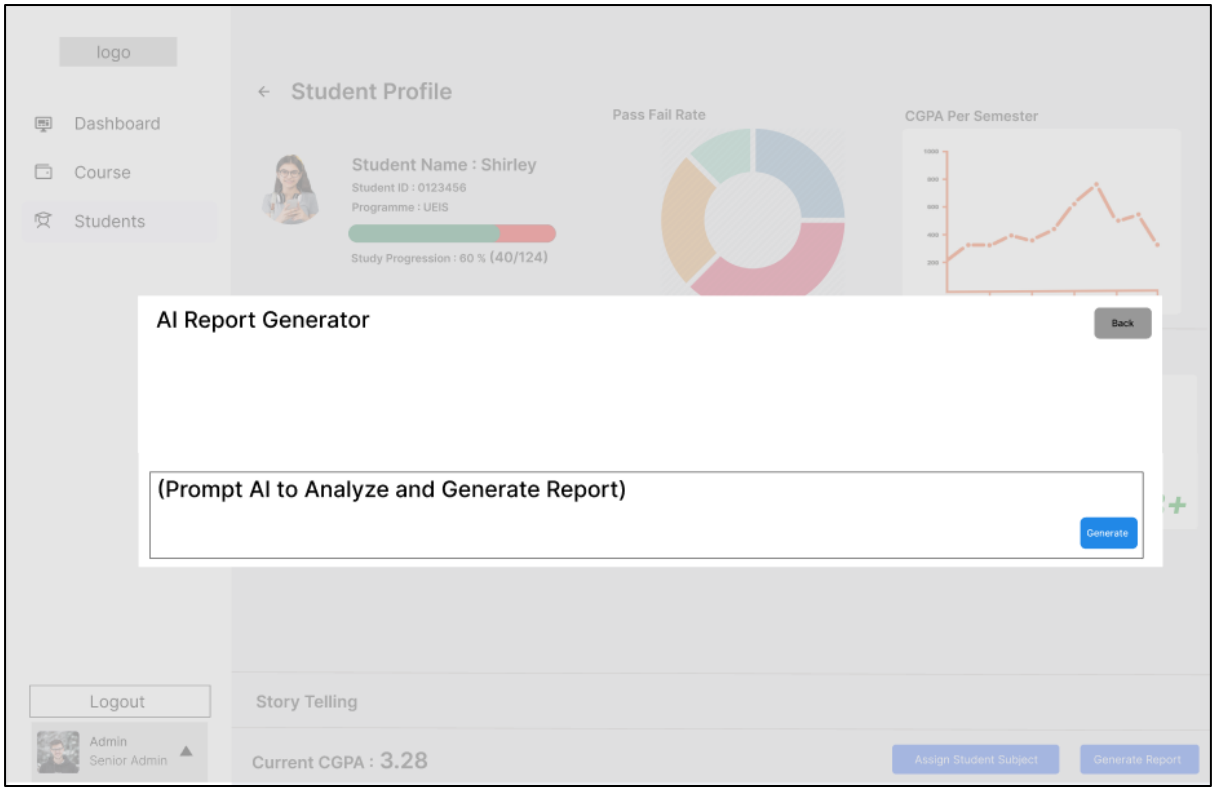
As for function of this page, the system provides a CGPA prediction function that allows user to look for future student performance (CGPA) by using Machine Learning.



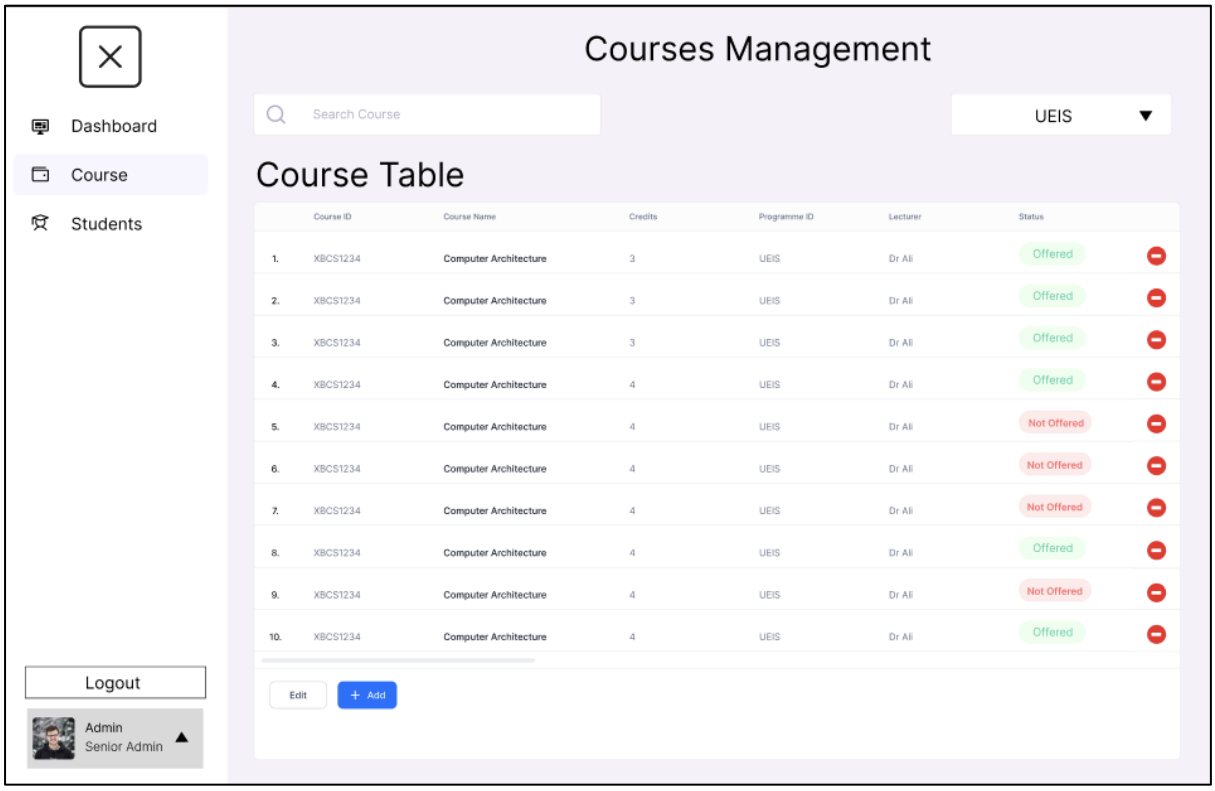
Besides that, the user can click the “Assign Student Subject Button” to manually assign the planned future courses by selecting offered courses with a checklist and selecting a semester.



Furthermore, the user can also use the report generation function to generate the student summary report through AI generation once the user clicked “Generate Report Button”.



3.6 Course Management Page

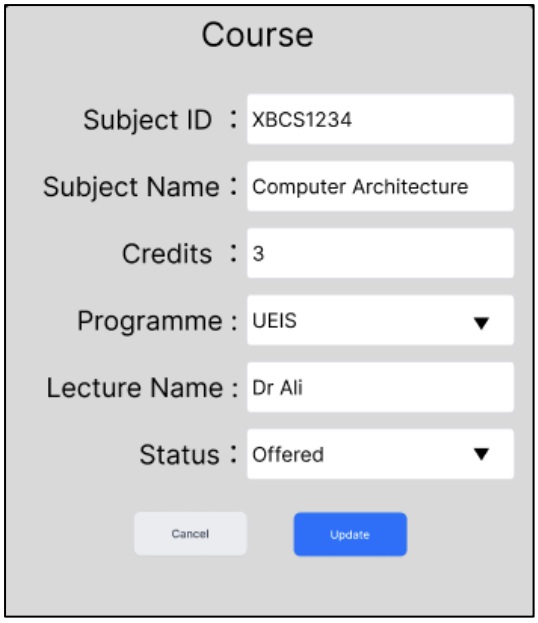


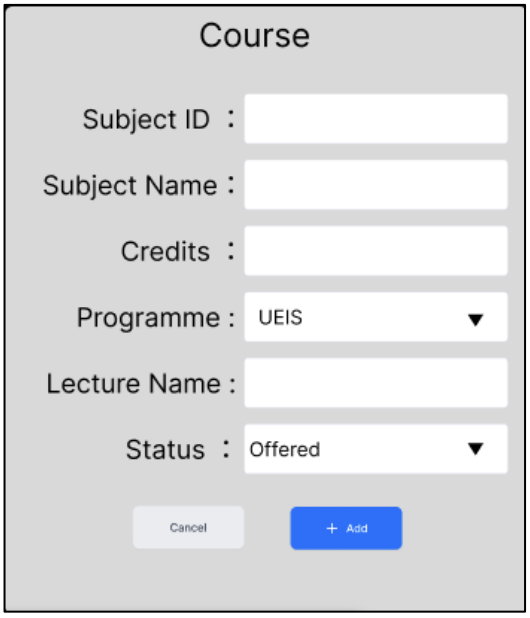
The figure above shows the course management page where it mainly displays the course list including the information subject ID, course name, course credits, programme ID, Lecture Name and course status for each course.

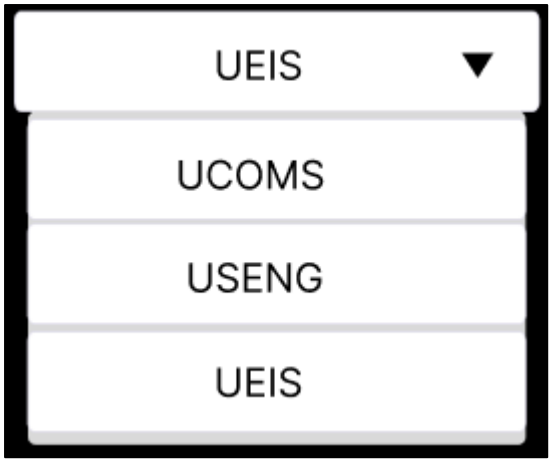
The system also provides a course filtering function that allow user to filter out the course needed. Besides that, user can add, remove and edit the courses to manage the courses.

After the user selects a course in the course table, the system will direct them to the course dashboard page.

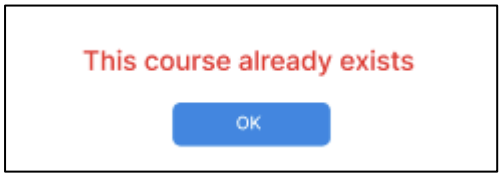
When the user need add or edit the courses, system will prompt user to input the new details of the course. Furthermore, user also can through choosing Programme ID to filter the courses that included in the programme. The figures below show the course details form and the dropbox detail that included in the form.





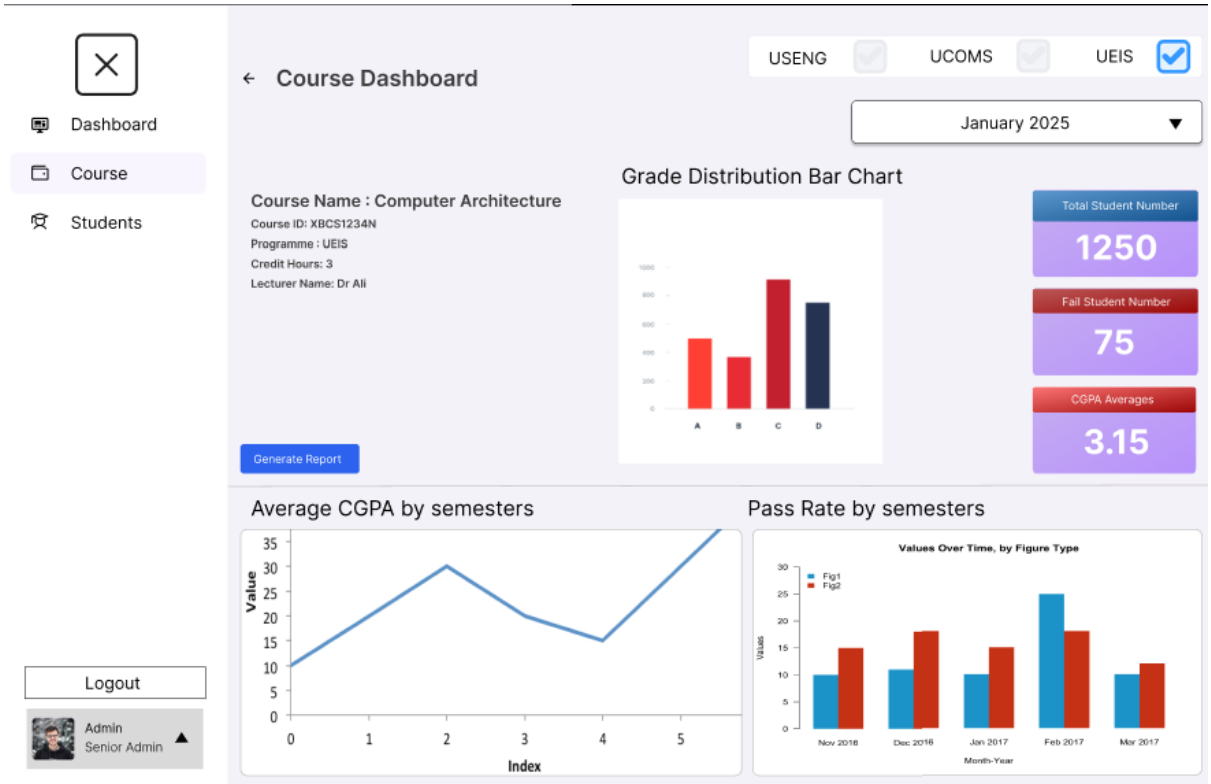


If the user submit courses that already exists form the course list. System will display an error message to notice user and return to the course management page without changing of the course list. The figure shows the error message alert.



Furthermore, user also can through choosing Programme ID to filter the courses that included in the programme.

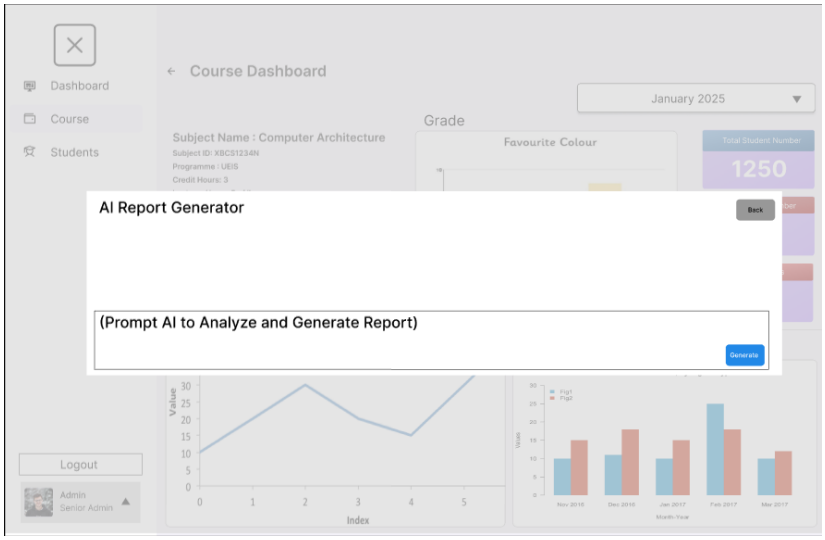
3.7 Course Dashboard Page



The course dashboard page displays general information about a specific course such as course name, course id, programme, credit hours and lecturer name. It displays the student's performance in a particular semester by showing the total failed student and the cgpa average, visualizing the grade distribution with a bar chart.

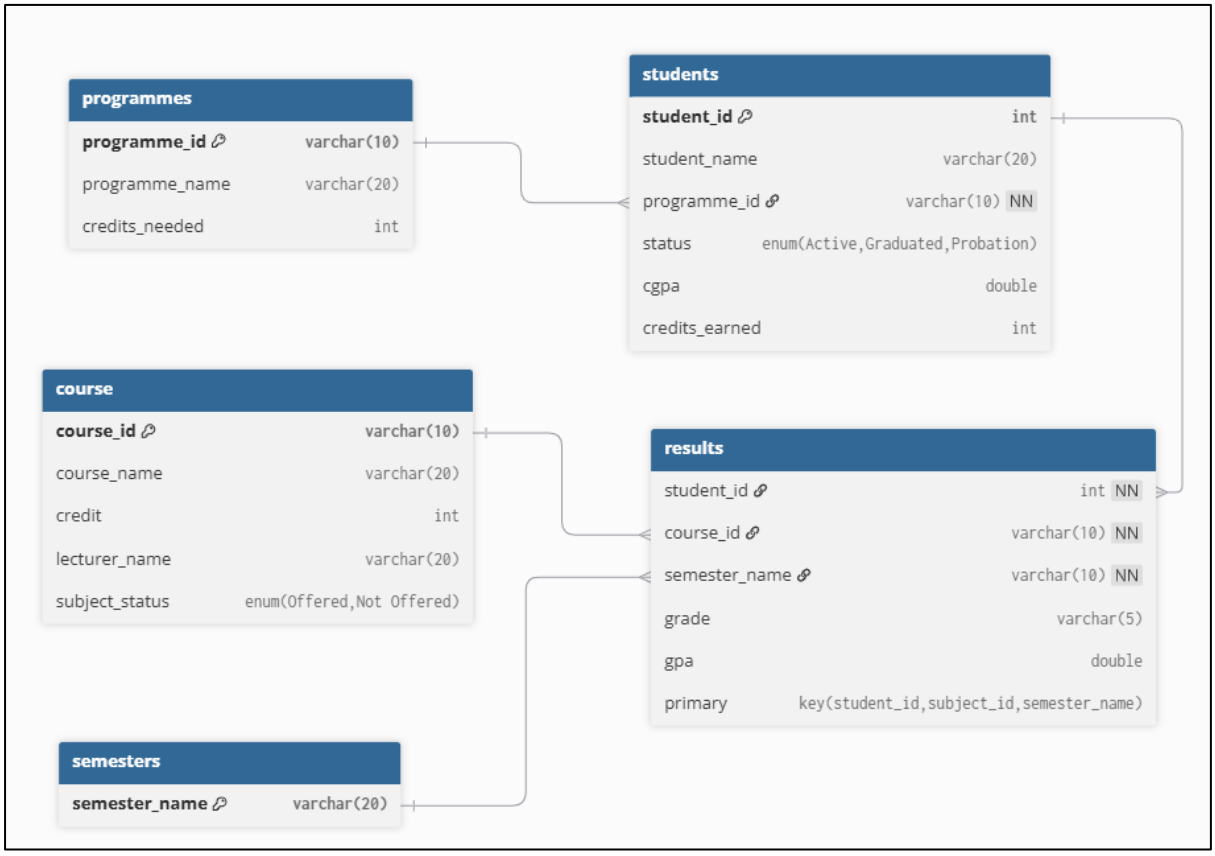
It also displays a general visualization of data such as the course's average cgpa by semesters and the course's passing rate by semesters.

The user can also prompt the ai report generator with the "Generate Report" button. The system will analyze and provide a storytelling of the data.



4.0 Database Design

4.1 Entity Relationship Diagram



4.2 Core Tables

Students

Captures student details including their status (active, graduated, or probation), academic performance, and enrollment in a programme.

student_id	student_name	programme_id	status	cgpa	credits_earned
0130001	Alice Lee	UCOMS	Active	3.45	90
0130002	John Tan	USENG	Graduated	3.85	125
0130003	Nurul Afiqah	UEIS	Probation	2.10	65

Programmes

Stores information about academic programmes such as Bachelor of IT, Software Engineering, etc., and the total credits required to graduate.

programme_id	programme_name	credits_needed
UCOMS	Bachelor of Computer Science	127
USENG	Bachelor of Software Engineering	127
UEIS	Bachelor of Enterprise Information System	127

Course

Holds data about subjects/modules including credit value, assigned lecturer, and current offering status.

course_id	course_name	credit	lecturer_name	course_status
XBCS2001N	Fundamentals of Programming	3	DrLim	Offered
XBCS2002N	Internet and Web Development	3	MsChen	Offered
XBCS2003N	Java Programming	3	MrWong	Not Offered

Results

Records students' academic performance per subject and semester, including grades and GPA.

student_id	course_id	semester	grade	gpa
0130001	XBCS2001N	January 2024	A-	3.67
0130001	XBCS2002N	May 2024	B+	3.33
0130002	XBCS2002N	January 2023	A	4.00
0130002	XBCS2003N	May 2023	A-	3.67
0130003	XBCS2001N	September 2024	C	2.00
0130003	XBCS2003N	January 2025	C+	2.33

Semester

semester_name
January 2024
May 2024
September 2024

4.3 Foreign Keys and Relationships

students.programme_id→programmes.programme_id
Each student is enrolled in one programme. A programme can have many students. One-to-many relationship.

results.student_id→students.student_id
Each grade record is associated with one student. A student can have multiple subject records.
One-to-many relationship.

results.course_id→course.course_id
Each grade record is associated with one subject. A subject can be taken by many students. One-to-many relationship.

results.semester_name → semester.semester_name

Each grade record is associated with one semester. One semester can have many subject records.

4.4 Normalization

This database follows Third Normal Form(3NF)

- 1NF: All attributes hold atomic values (no repeating groups)
- 2NF: All non-key field depend on the full primary key
- 3NF: No transitive dependency exists

This design avoids redundancy, supports scalability, and ensures data consistency across the system.

This database design also follows the Boyce-Codd Normal Form(BCNF) where for every functional dependency X→Y, X is a primary key.

4.5 Schema Explanation

Programme Table

Field	Type	Description
programme_id	varchar(20)	Unique ID for each academic programme
programme_name	varchar(10)	Name of the programme
credits_needed	int	Total credits required to graduate

Student Table

Field	Type	Description
student_id	int	Unique student identifier
student_name	varchar(20)	Full name of the student
programme_id	int	Foreign key linked to programme table
status	enum	Academic status (Active, Graduated, Probation)
cgpa	double	Cumulative GPA
credits_earned	Int	Total credits earned by student

Course Table

Field	Type	Description
course_id	int	Unique student identifier
course_name	varchar(20)	Full name of the student
credit	int	Credit weight of the subject
lecturer_name	varchar(10)	Name of the assigned lecturer
course_status	enum	Indicates if the subject is currently offered

Results Table

Field	Type	Description
student_id	int	Foreign key linked to student table
course_id	varchar(10)	Foreign key linked to subject table
semester	varchar(10)	Semester when subject was taken
grade	varchar(5)	Grade awarded (e.g. A, B+, C-)
gpa	double	GPA contribution for that subject

Semester Table

Field	Type	Description
semester_name	varchar(20)	Unique name for the semester

5.0 References

Chatti, M. A., Dyckhoff, A. L., Schroeder, U., & Thüs, H. (2012). *A reference model for learning analytics*.

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